

ORIENT AI

A research-driven investigation into vocational disorientation and a Machine Learning solution to fix it.

TEAM 7

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A SYSTEMIC CRISIS, NOT A PERSONAL ONE

This project began through informal conversations with students at the Politechnika University of Lodz.

- Across different faculties, a common pattern emerged: most students felt lost, overwhelmed, and disoriented when choosing their degree.

The transition from high school to university was described as:

Stressful

Uncertain

Blind
decisions

Many students choose their degree due to:

External pressure
(family, expectations)

Lack of
guidance

Process of
elimination

We identified a systemic failure in career guidance



VALIDATING THE PROBLEM



Is This Still Happening today?

This is not a past issue, it is a current and ongoing problem



VALIDATING THE PROBLEM

To validate our hypothesis, we reached out to current High School students in Spain and the Polytechnic University. Their feedback was clear:

- They experience the same confusion and frustration
- They feel unsupported by traditional guidance systems
- Despite digitalization, career orientation methods:
 - Have not evolved
 - Do not adapt to students' real needs

This is not a past issue, it is a current and ongoing problem



FROM INSIGHT TO DATA

To move from observation to data, we created a survey:

Distributed:

Via Google Forms

Target:

High School students currently choosing their future

Shared:

Through our former high schools in Spain

Delivered:

Directly in classrooms via teachers and counselors

Ensured a reliable and relevant sample

SURVEY

Main objectives of the Survey:



**Measure
Decision
Fatigue**

How overwhelmed
students feel with
current tests



**Assess Trust in
Guidance Systems**

Do existing tools
reflect their real
interests?



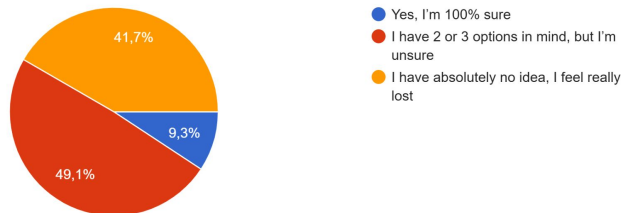
**Identify the
"Satisfaction
Gap"**

Fear of choosing
something that
doesn't make
them happy

SURVEY RESULTS: 108 STUDENTS

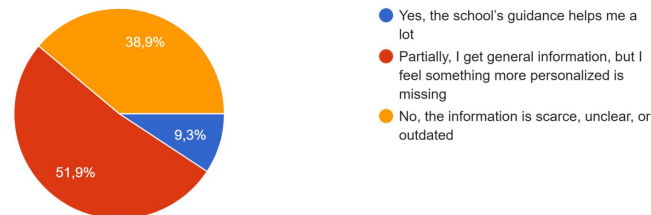
Right now, do you have a clear idea of what you'll study after high school?

108 respuestas



Do you feel you receive enough useful, clear, and personalized information at your school about the different degrees?

108 respuestas



When you think about choosing your major or program, what is your main feeling?

108 respuestas



SURVEY RESULTS: 108 STUDENTS

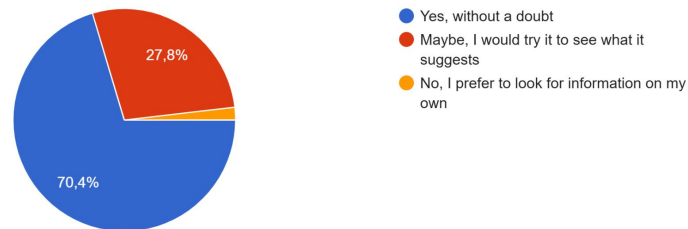
KEY FINDINGS

- 66% report high stress levels when choosing a degree
- 70% would use a mobile app



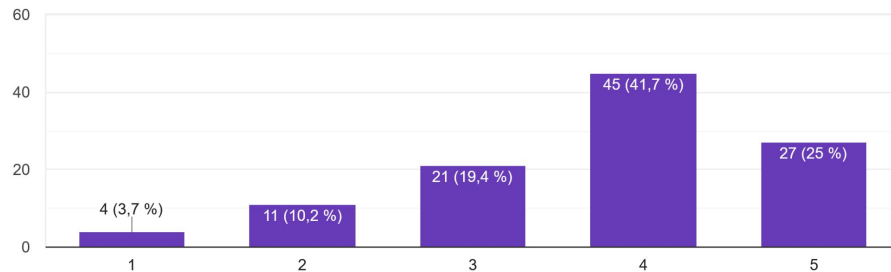
If there were a free mobile app that analyzed your real interests and recommended degrees that perfectly fit you, would you use it?

108 respuestas



From 1 (low) to 5 (high), how much stress do you feel about making this decision?

108 respuestas

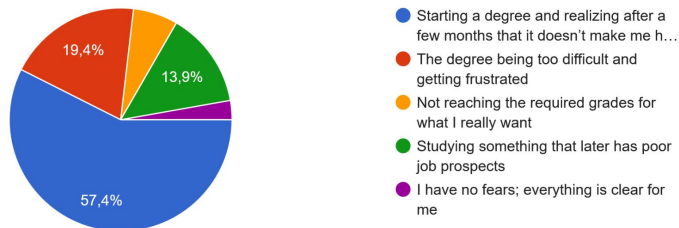


SURVEY RESULTS: 108 STUDENTS

Core Fear Driving Decisions

What is your biggest fear when it comes to deciding your academic future?

108 respuestas

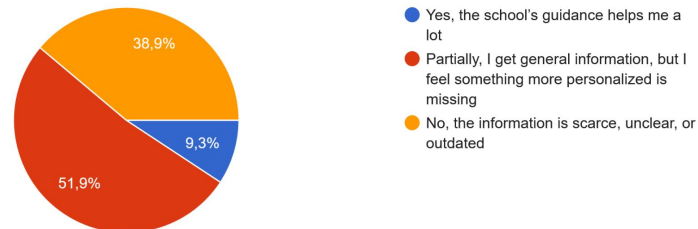


The problem is not only choosing, it is choosing "wrong"

Failure of Current Systems

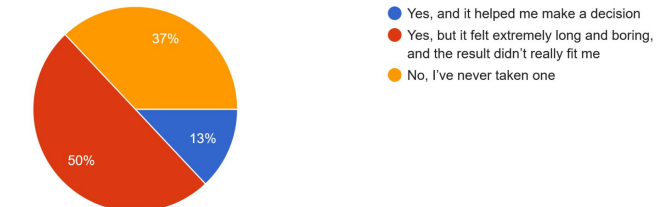
Do you feel you receive enough useful, clear, and personalized information at your school about the different degrees?

108 respuestas



Have you ever taken a classic career guidance test?

108 respuestas



Current tools generate decision fatigue instead of clarity

STRONG DEMAND FOR A NEW SOLUTION



70.1% would use an app that uses AI to recommend a degree and predict a 'Satisfaction Index' (e.g., 85% likelihood of happiness/not dropping out).



76.6% find AI-based satisfaction prediction “incredible”



90.7% prefer a dynamic, fast experience

We can see a clear demand for a personalized, intelligent, and efficient tool

STATE OF THE ART: 2025-2026

Pymetrics (Neuroscience-Based)

Method: Uses neuroscience games and AI to assess cognitive and emotional traits.

Goal: Matches users with roles where similar profiles have succeeded.

Kickresume (AI Career Coach)

Method: Analyzes resumes using LLMs (Large Language Models).

Goal: Identifies skill gaps and suggests career paths for professional transitions.

AlCareerCoach.org (Conversational Mentor)

Method: A free, chatbot-style tool by CareerVillage.

Goal: Acts as a mentor for students to explore interests through dialogue.

Steppingblocks (Big Data Insights)

Method: Educational data analytics using "Big Data."

Goal: Shows real-world career paths based on the outcomes of millions of professionals.

IBM Watson Career Coach (Corporate NLP)

Method: Utilizes Natural Language Processing (NLP).

Goal: Helps employees find internal growth opportunities aligned with their skills.

WHY TRADITIONAL GUIDANCE FAILS

Inflexibility



Market
Disconnect



Wellbeing
Blind Spot



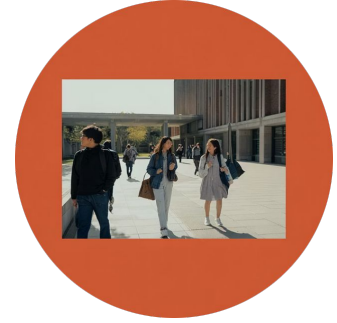
THE USERS/STAKEHOLDERS: WHO WE'RE BUILDING FOR



Primary: Students



Secondary: Counselors &
Parents



Tertiary: Universities &
Employers

THREE MAIN PROBLEMS, THAT WE WILL SOLVE...

1



Decision Fatigue

2



Rigid Rule-Based Models

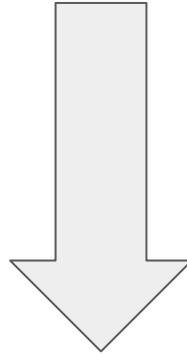
3



Aptitude $\neq$ Satisfaction

PROBLEM TO BE SOLVED

The **lack of** vocational **guidance tools** that integrate multi-stage quantitative data analysis and the prediction of subjective satisfaction, can **lead to dropout** or **poor career choices** based on incomplete information or decision fatigue.



OUR SOLUTION: ORIENT AI

ORIENT AI



Data Input optimization

Cascading surveys :

Level 1: Scope → Level 2: Specialty

Nonlinear Model

multi-class classification model
can identify profiles traditional
tests miss

Career + Score

Not only suggests a career path
but also quantifies the probability
of the user feeling satisfied



ANY QUESTIONS ?